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France
Europe

VISA STATUS

Authorized to work in France immediately — eligible for CDI

Open to relocation across Europe, Singapore, Dubai.

PROFESSIONAL INTERESTS

Naval architecture

Hydrodynamics & seakeeping

Offshore engineering & operations

Structural/Mechanical analysis

Project coordination & interface management

Offshore energy (oil and gas, offshore wind)

SOFTWARE SKILLS

MAAT hydro++

AutoCAD

Abaqus/CAE

SolidWorks

OrcaFlex

NEMOH

STAR-CCM+

Python & MATLAB

MS Office Suite

LANGUAGES

English
Fluent

French
Intermediate

Tamil
Fluent

Vivek Chander MUNIGAN

Marine Technology Engineer | Naval Architect | Hydrodynamics (Ocean Engineering)

Marine Technology Engineer (BEng Naval Architecture & Offshore Engineering, MSc Hydrodynamics) with 4+ years' combined experience across ship design, ship repair/site work, and maritime compliance documentation. I combine structural and hydrodynamic analysis (FEA) with clear technical documentation and coordination with owners, shipyards, and classification societies.

WORK EXPERIENCE

03/2025 - 08/2025 — ENSTA Bretagne & France Énergies Marines, Brest, France

R&D Engineering Intern – Dynamic Submarine Power Cables

- Developed a **coupled MATLAB–Abaqus** thermo-mechanical model for dynamic submarine power cables, reducing manual preprocessing time by 40 %.
- Ran **FE simulations** (bending / tension) with **mesh sensitivity** and strength checks to validate **cable behavior**.
- Built a **temperature-dependent** material database and automated simulation workflows using **Python**, improving analysis throughout.
- Authored clear **technical reports** and **calculation notes** for project stakeholders.

08/2020 - 08/2023 — Sea Sentinels (Remote), Singapore

IHM Project Coordinator | Ship Recycling Consultant

- Coordinated interfaces with **shipowners, shipyards, and class** to deliver compliant documentation under **tight deadlines** (equipment lists, SDS, MDOC/SDOC, IHM).
- Prepared and reviewed **IHM survey documentation** for a wide range of vessel types (tankers, bulk carriers, cargo vessels, and others) in line with **EU and class requirements** (***ABS, LR, DNV, BV***).
- Supported **onboard IHM surveys**, including **night surveys during short port calls**, ensuring complete hazardous materials mapping.
- Served as client-side **HSE representative** at Alang ship recycling yard (Gujarat), verifying adherence to agreed **procedures and safety rules** and supporting incident-free progress.

06/2017 - 05/2018 — GreenSHIP Research and Technology, Chennai, India

Naval Architect

- Pioneered the design of a barge, **conducting stability, resistance and powering calculations** that validated performance metrics and informed the **structural design approach**.
- Engineered a **cost-optimized material solution** through **comparative analysis**, achieving approximately 15% cost reduction while meeting all **classification society standards**.
- Contributed to strategic **business development** by representing technical capabilities in **client meetings** and supporting the creation of marketing collateral that **won new projects**.

11/2016 - 04/2017 — Care Marine & Allied Services, Mumbai, India

Site Engineer

- Ensured **no major non-conformities** reported by performing meticulous inspections and providing direct supervision of ship **repair activities**, guaranteeing **strict compliance with all safety and quality standards**.
- Coordinated the commissioning and readiness of a **temporary firefighting system for naval vessel INS Skathi A57**, ensuring all systems were **operational and certified** for duty.

08/2016 - 11/2016 — ADONIA Design Engineering & Consulting LLP, Mumbai, India

Trainee Naval Architect

- Executed **statutory plan** development for vessel flag conversions, delivering compliant **GA, LSS, LSA, FCP, and GMDSS** drawings that secured expedited **IRS approval**.
- Navigated international regulations (**SOLAS, MARPOL**) to ensure all documentation met **flag and class requirements**, resulting in **on-time regulatory approval.**

HOBBIES

- **Sports**
- **Cooking**

CORE SKILLS

Hydrodynamics and wave loading concepts with seakeeping basics (RAOs/BEM mindset); numerical analysis and verification mindset; method selection and structured engineering calculations; technical reporting and decision-ready deliverables; interface coordination with multidisciplinary stakeholders; planning and delivery discipline under time constraints; engineering software and automation (HydroSTAR/NEMOH, MATLAB, Python; FEA exposure Abaqus/CAE); fluent English and comfortable in international environments.

ACADEMIC PROJECTS

Stability of fishing vessel - MAAT Hydro++

- Performed intact and damaged stability analysis, strengthening expertise in **structural safety and verification methods**.

Mooring of tankers - DEEPLINES

- Designed and analysed tanker mooring system using Deeplines, evaluating hydrodynamic responses and line behaviour — knowledge applicable to **flowline-soil and mooring-line interactions**.

Water Waves and Sea State Modelling - MATLAB

- **Modelling and analysis** of waves in towing tank basin.

Seakeeping lab work using the BEM solver NEMOH - MATLAB

- RAOs computation, Mesh convergence analysis, Diffraction-radiation force calculation.

Numerical Simulation of Turbulent Flow Over a Periodic Hill - STAR-CCM+

- Conducted CFD analysis using STAR-CCM+ to evaluate turbulence models and mesh configurations for simulating 2D turbulent flow over a periodic hill.

Particle trajectories and mass transport of water-waves MATLAB

- Simulated fluid particle trajectories and estimated the corresponding drift speed.

EDUCATION

09/2023 - 08/2025 — École Centrale de Nantes, Nantes, France

Masters 1 & 2 in Marine Technology - Hydrodynamics for Ocean Engineering

Focus on ship hydrodynamics, stability, structural analysis and numerical modelling of offshore systems.

Key Topics:

Continuum Mechanics, Experimental Hydrodynamics, Hydrodynamics R&D, Numerical Hydrodynamics, Ship Stability, Water Waves & Sea State Modelling, CFD / viscous flows and Mooring Analysis.

08/2012 - 05/2016 — AMET University, Chennai, TN, India

Bachelor Of Engineering In Naval Architecture And Offshore Engineering

Key Topics:

Ship Design, Ship Resistance and Propulsion, Fluid Mechanics, Analysis of Structures (FPSO/FSO), Design of Offshore Structures, Fundamentals of Naval Architecture, Ship Stability and Material Science.